

Exploring the data

DATA LITERACY CASE STUDY: REMOTE WORKING ANALYSIS

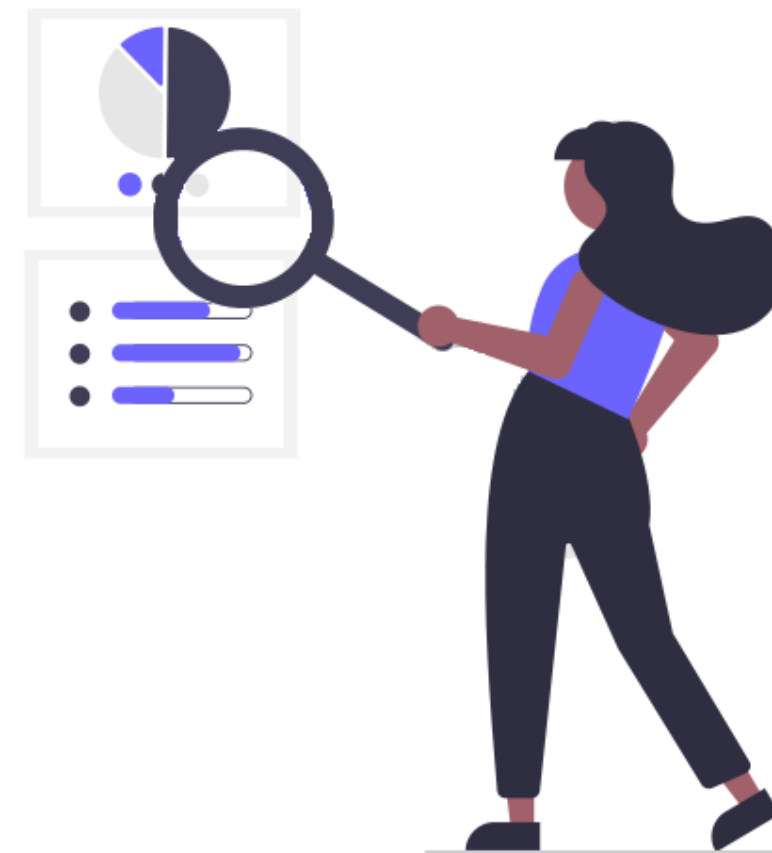


Maarten Van den Broeck

Senior Content Developer at DataCamp

Exploratory data analysis

- Why?
 - Assessing the main characteristics of the data
 - Finding relationships, patterns, or groups
 - Suggesting questions or hypotheses for future analysis
- How?
 - Describing the data numerically
 - Visualizing the data



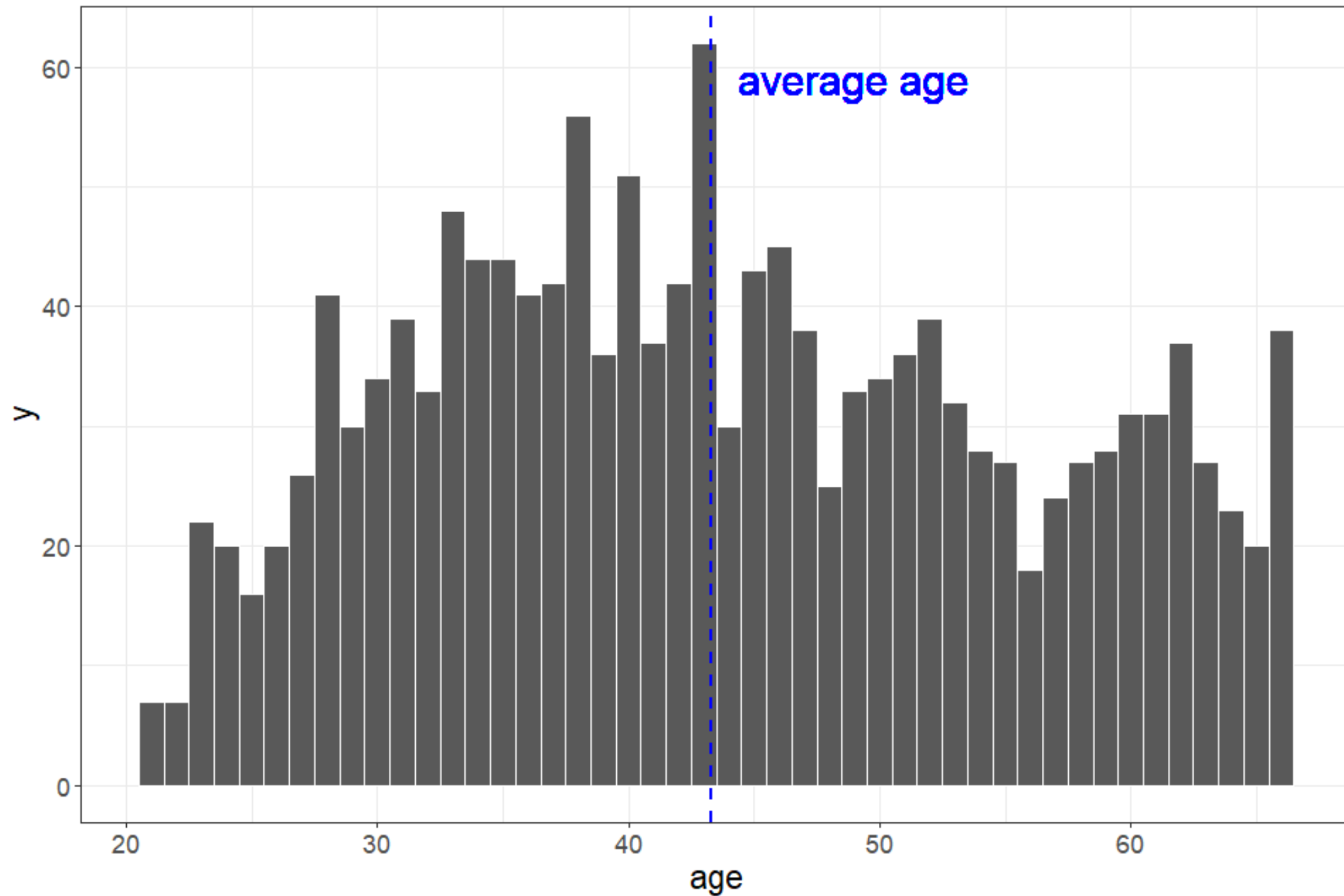
Example: age characteristics

- What are the main age characteristics of the respondents?
 - Average/median age: 43
 - Youngest: 21
 - Oldest: 66

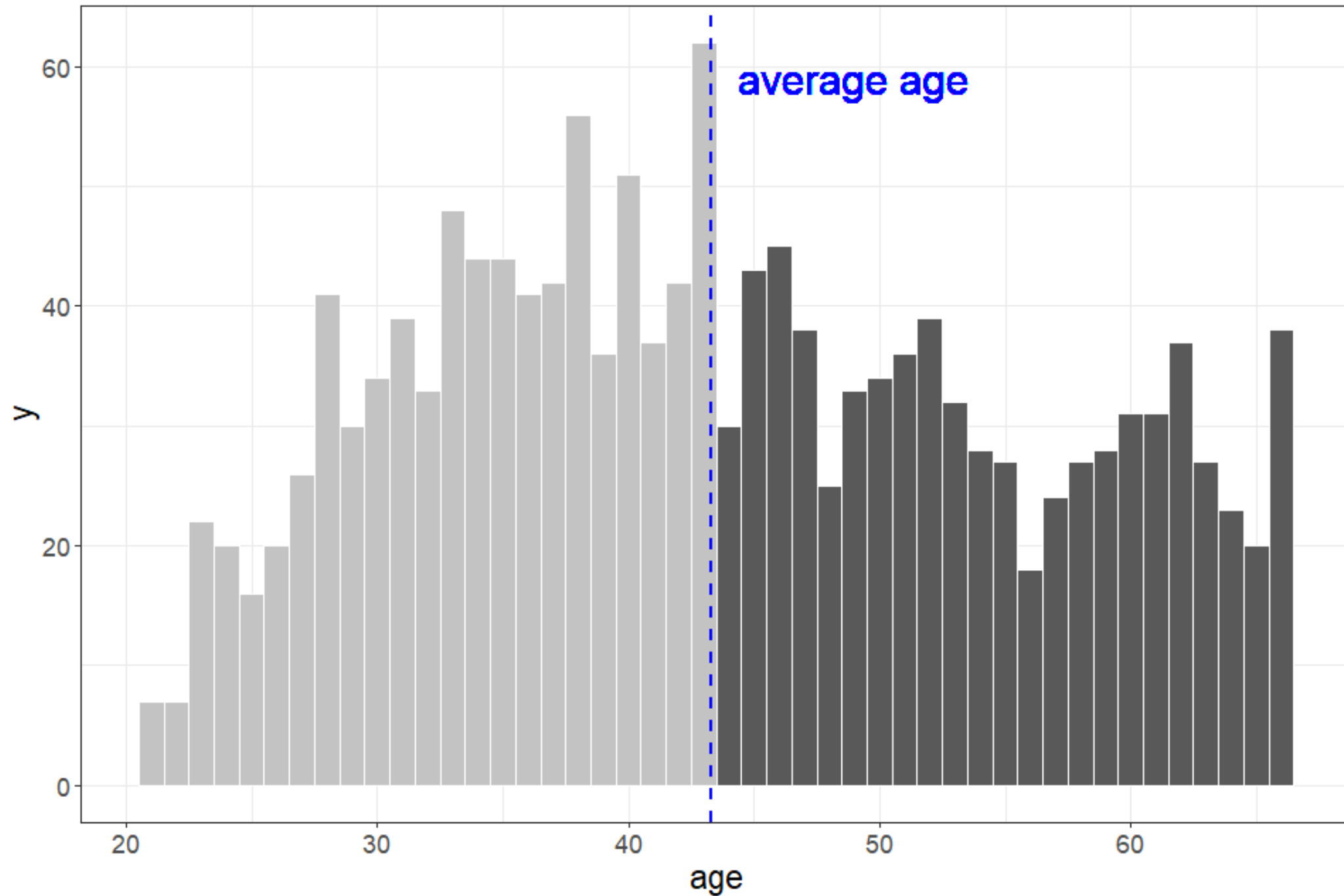
Descriptive statistics of age

	Age
Mean	43.77
Std.dev	11.84
Min	21.00
Median	43.00
Max	66.00
N valid	1512

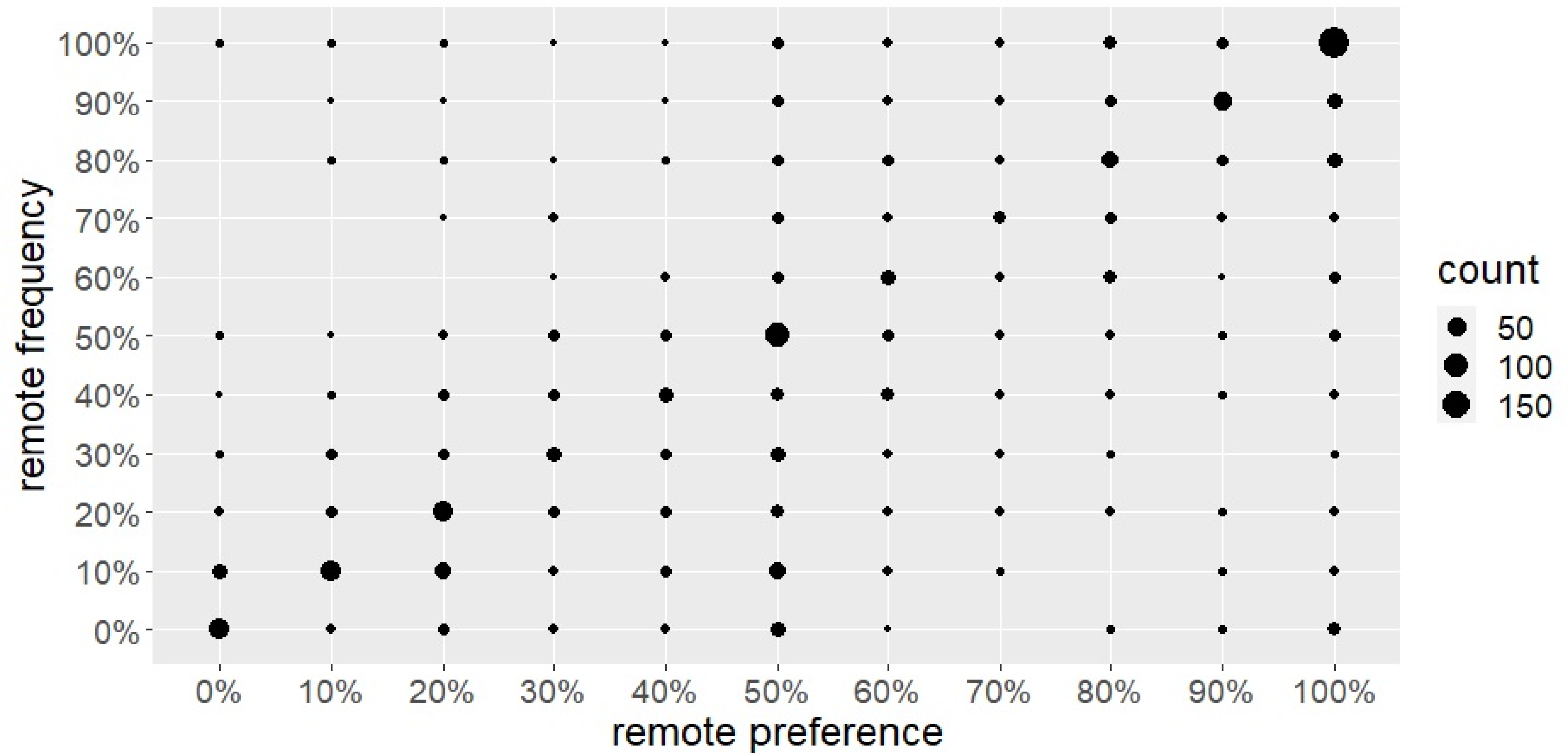
Are all age groups represented?



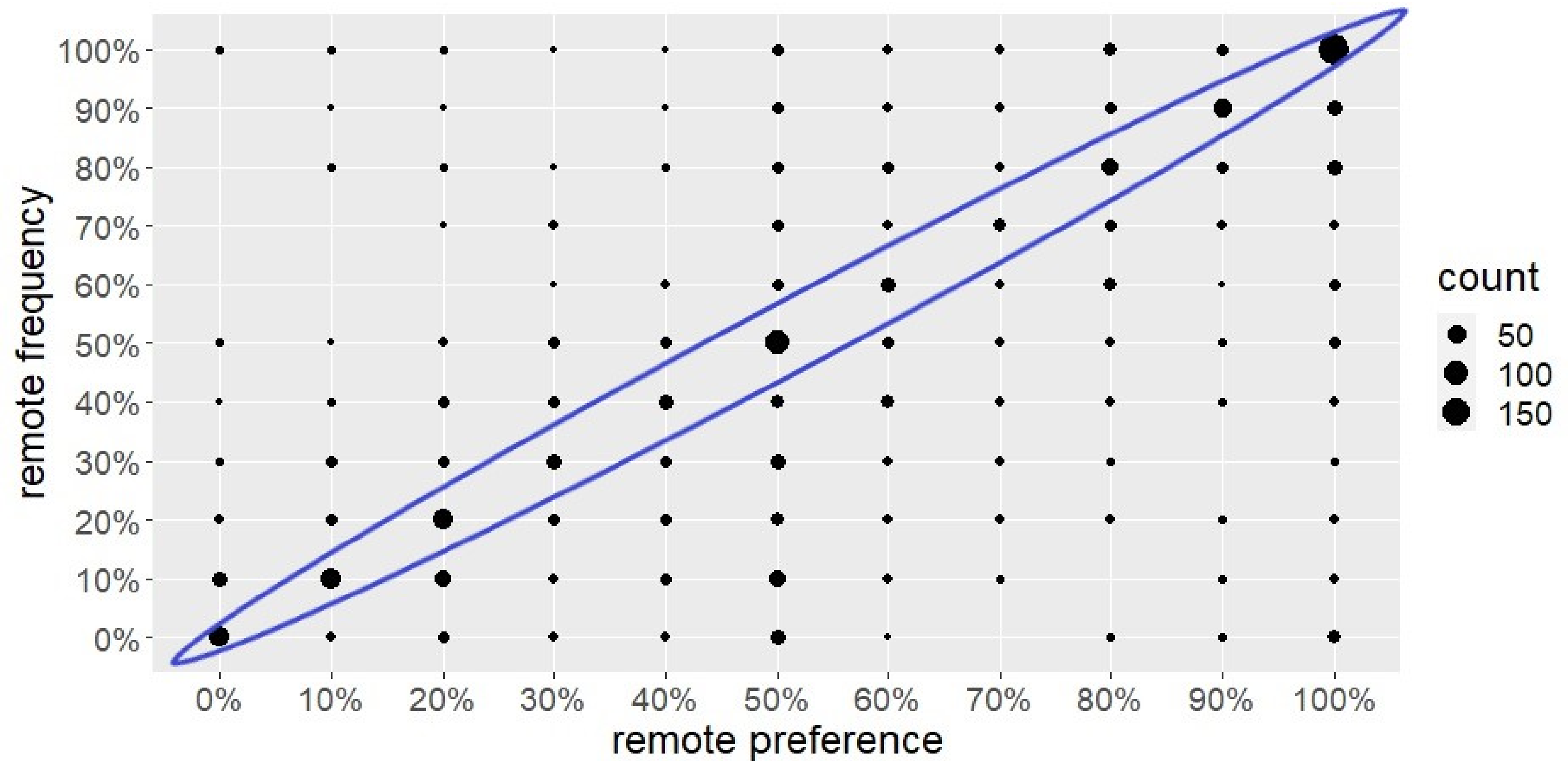
Are all age groups represented?



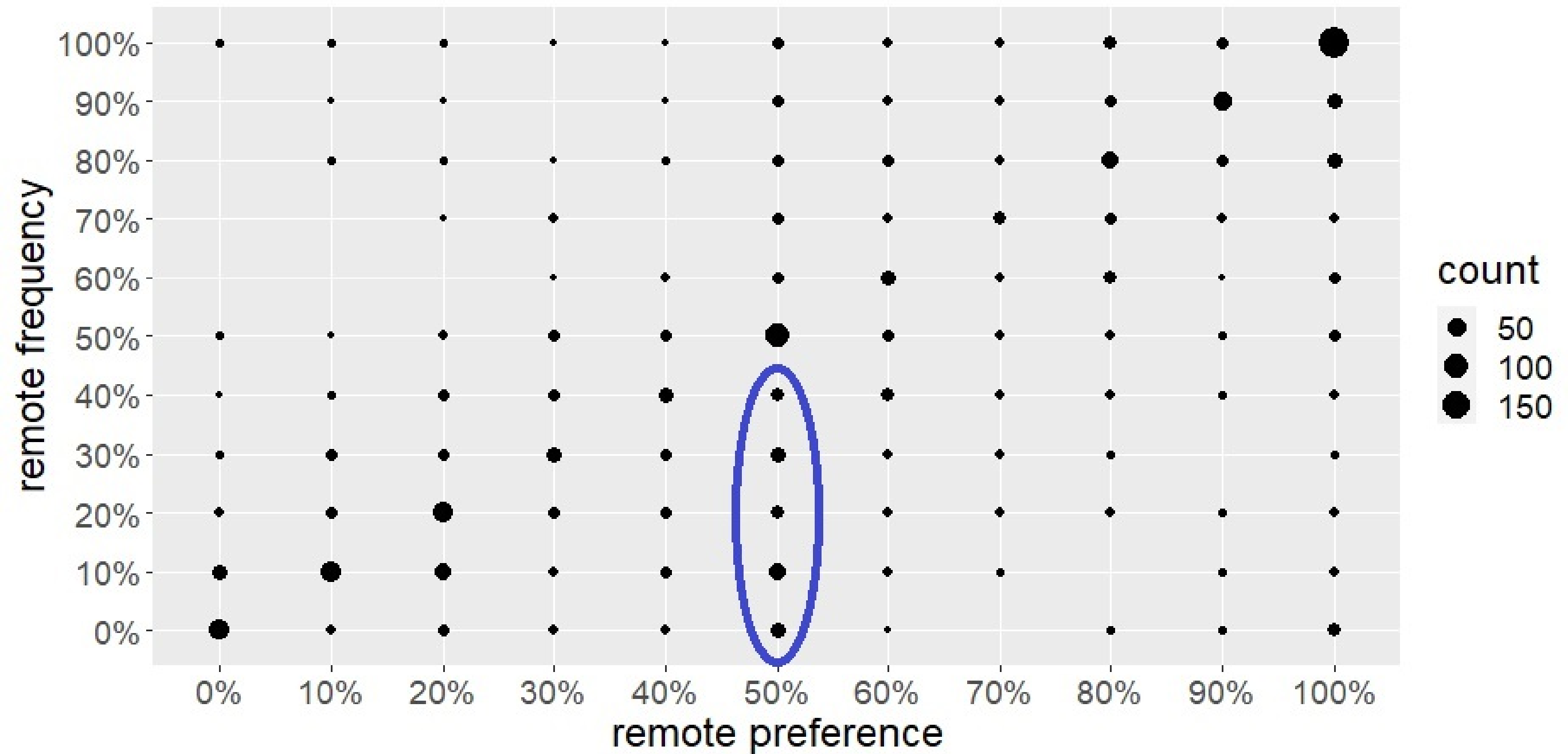
Example: remote frequency vs. preference



Example: remote frequency vs. preference



Example: remote frequency vs. preference



Let's practice!

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Getting insights from the data

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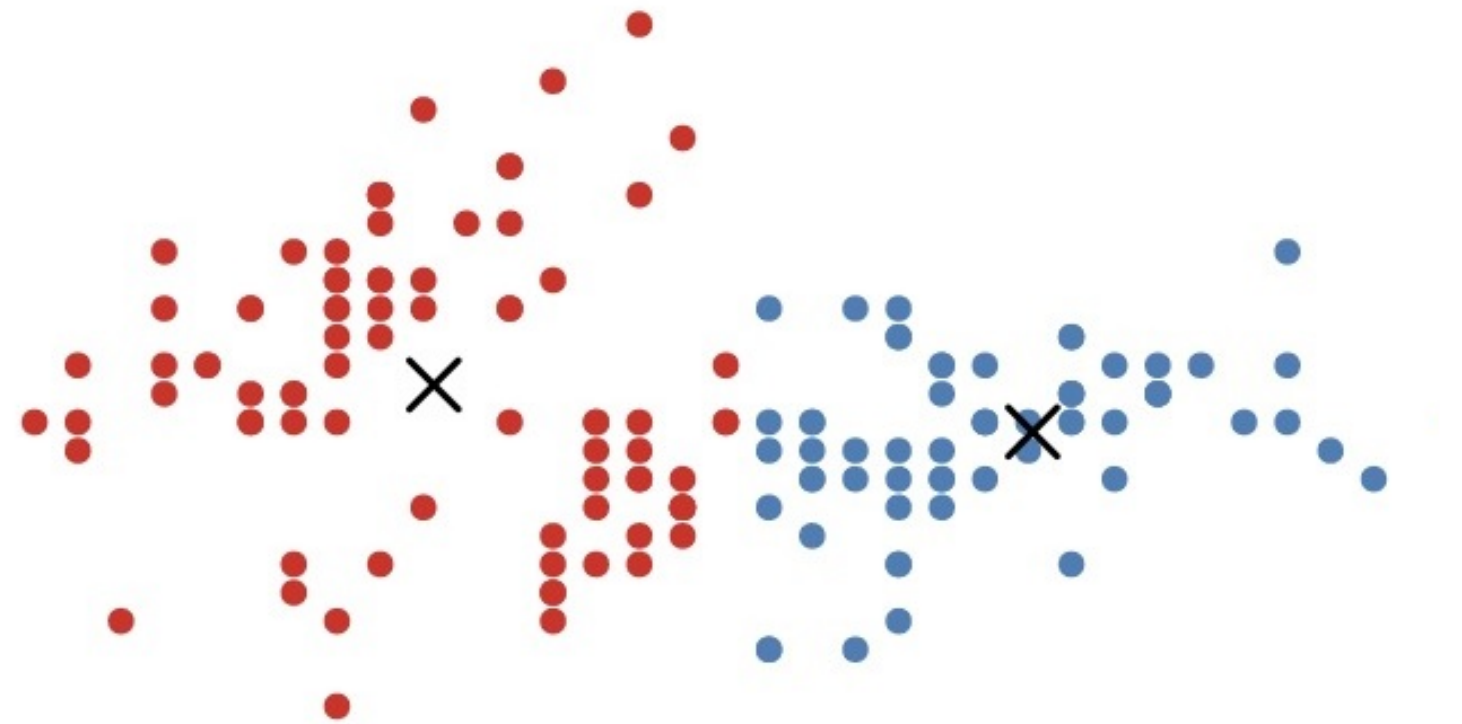


Maarten Van den Broeck

Senior Content Developer at DataCamp

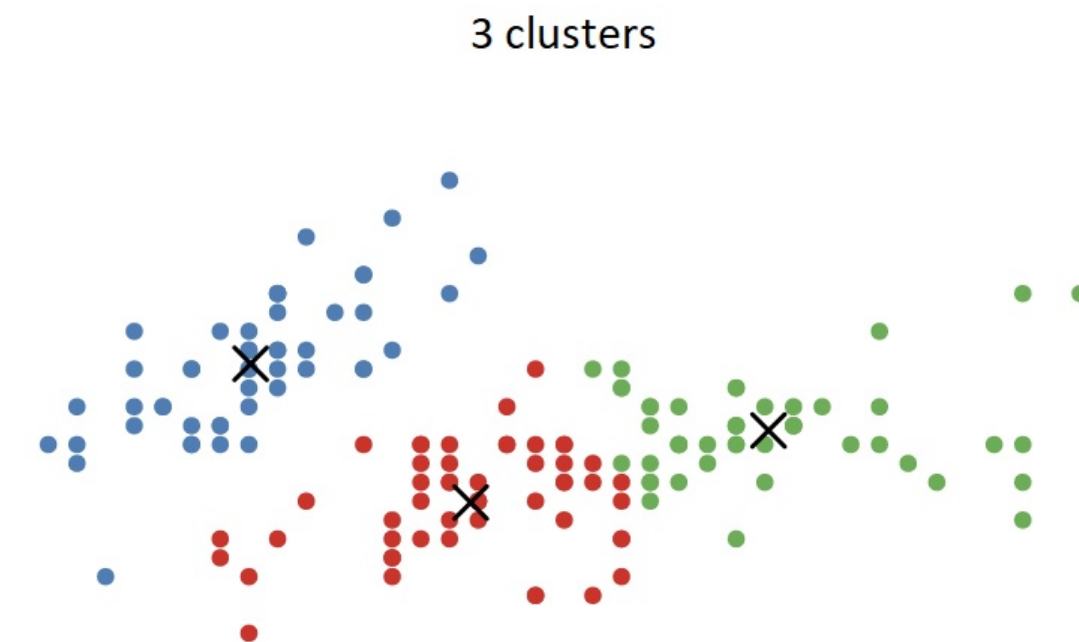
Cluster analysis

- Common descriptive and exploratory technique
- Goal: find naturally occurring groups in the data
- Possible applications:
 - Customer segmentation
 - Making a classification
 - Identifying subgroups
- Two main steps:
 - Finding the optimal number of groups
 - Investigating the characteristics of each



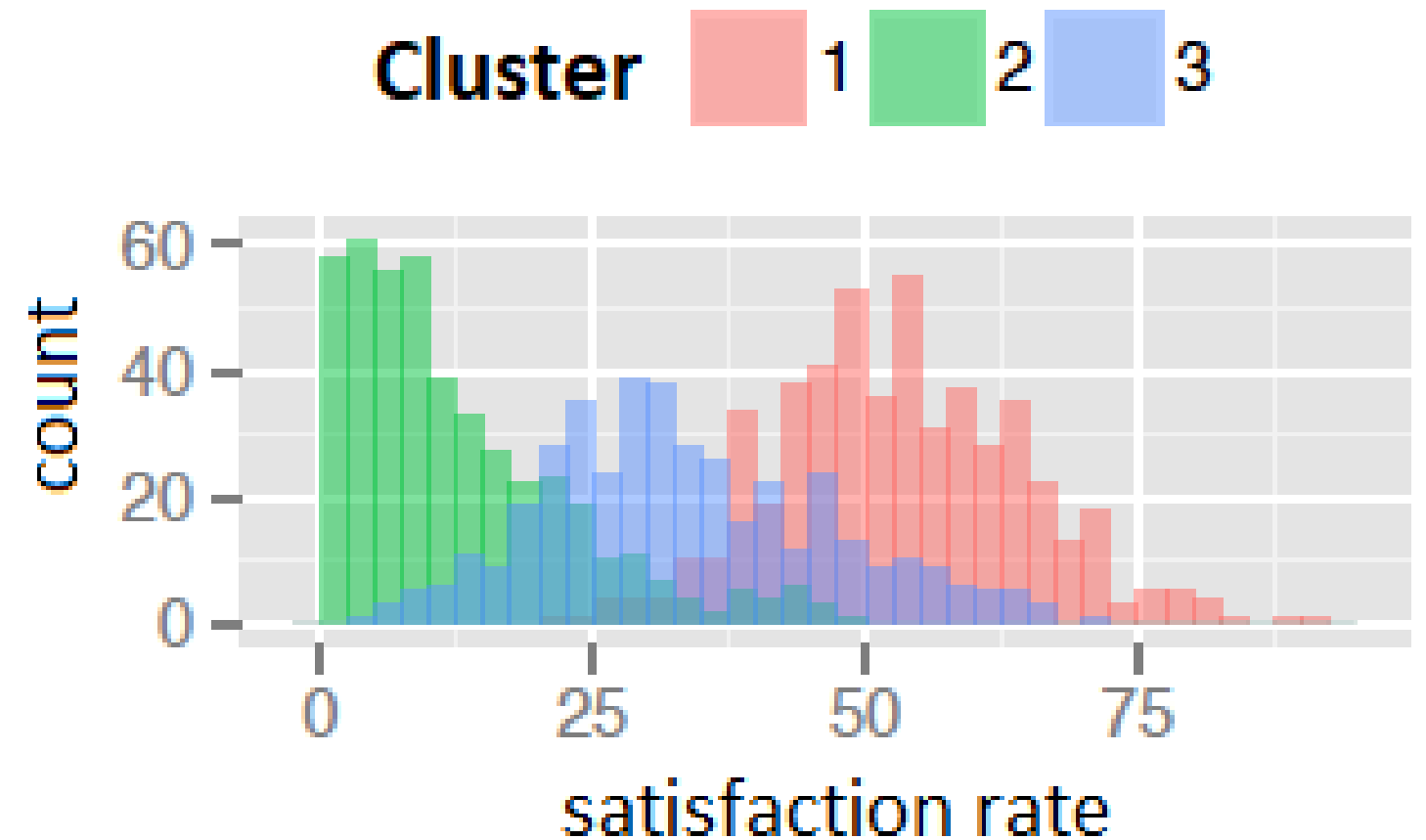
Finding the optimal number of groups

- Calculation of the optimal solution
- Domain knowledge: experts/business
- Not always an exact solution!



Characteristics of the groups

- How are they different?
- Common characteristics between a subset of groups
- Variables of interest



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Visualizing and communicating your insights

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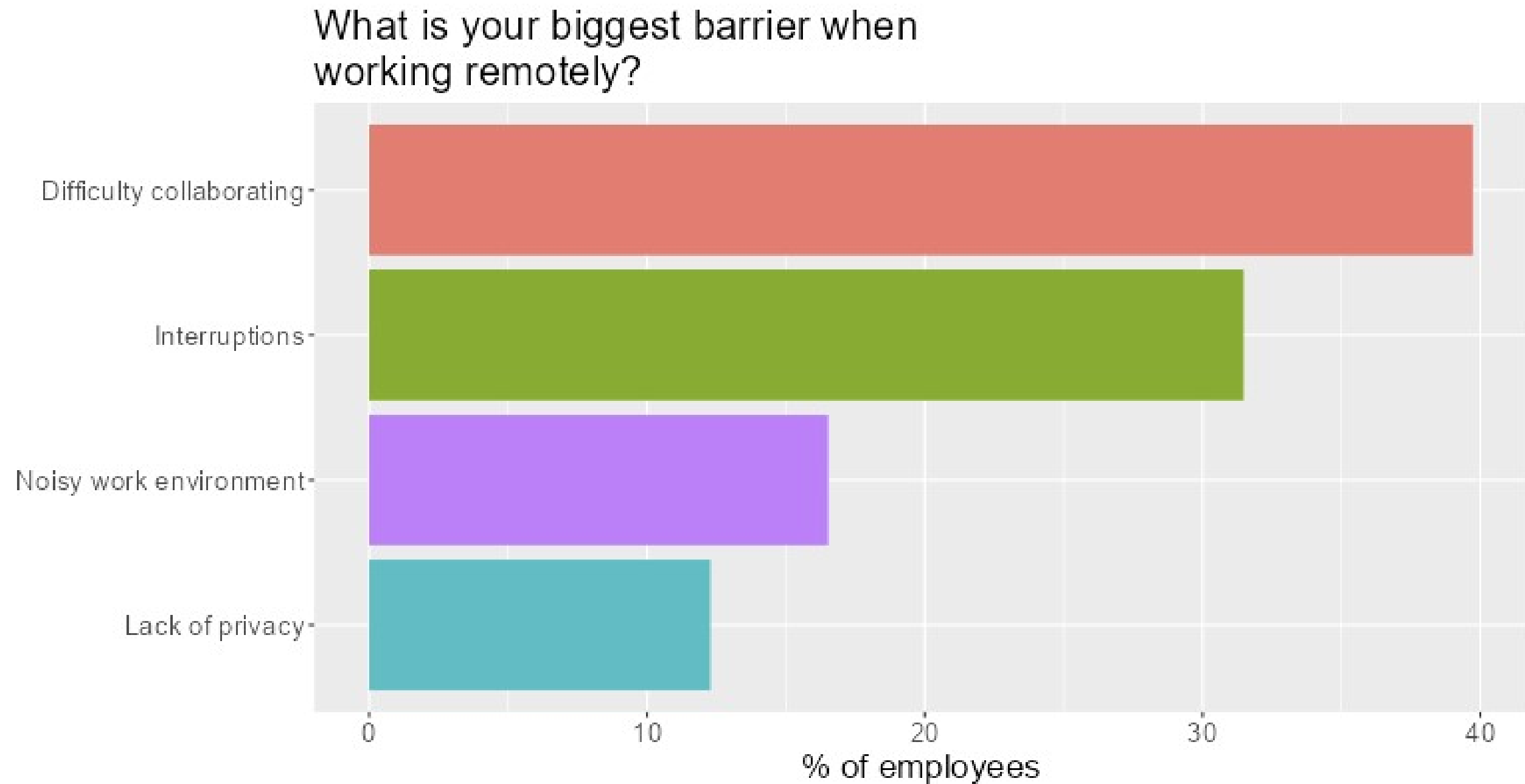
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A picture tells a thousand words

What is your biggest barrier while working remotely?	
1. Difficulty collaborating	40%
2. Interruptions	31.5%
3. Noisy work environment	16%
4. Lack of privacy	12.5%

A picture tells a thousand words



Things to consider when communicating your insights

- Choose the right format
- Use a narrative structure
- Support insights with visualizations



Choose the right format

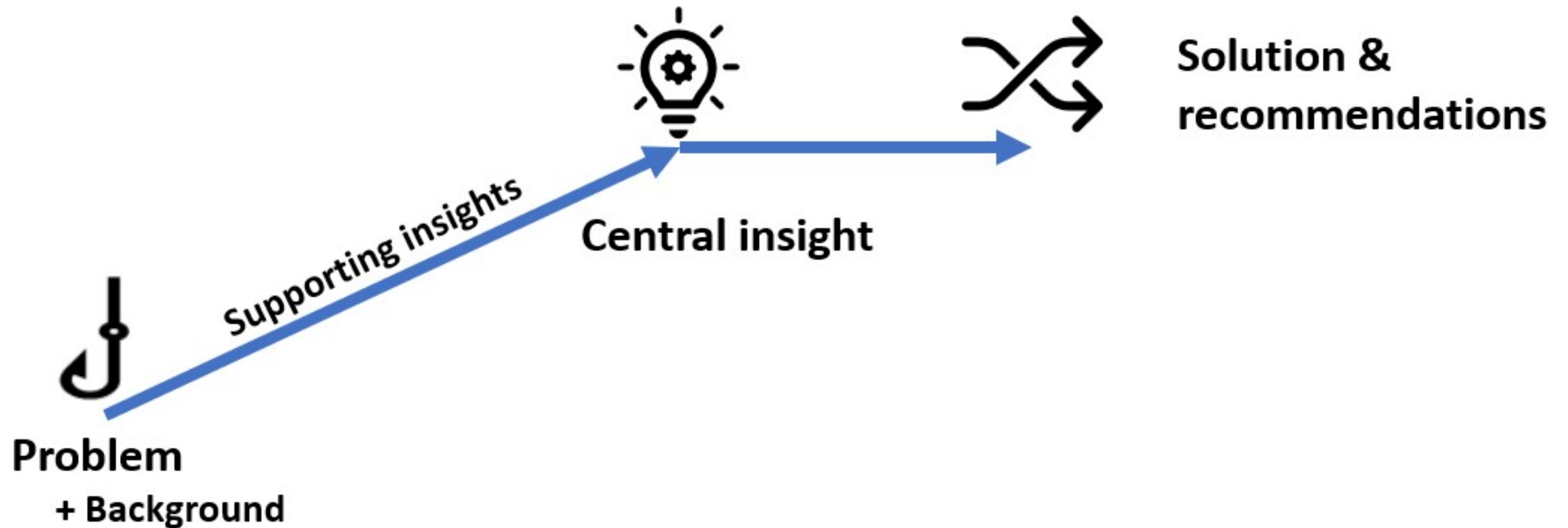
Final report/presentation

- For management/business
- Meant to communicate actionable insights
- Lots of visuals

Technical documentation

- For data specialists
- Meant to allow reproduction or follow-up analysis
- Detailed explanation of the methodology

Use a narrative structure



Support insights with visualizations

- Helps to understand complex information at a glance
- Makes it easier to understand and remember insights identified in the data
- Focuses attention on key aspects of your results



Let's practice!

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Wrap-up

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Maarten Van den Broeck

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Recap

Chapter 1: Getting to Know the Case Study

- Learned about the case
- Defined analytical questions
- Planned the analysis strategy

Chapter 2: Investigating Remote Working Habits

- Explored the data
- Performed cluster analysis to solve the analytical questions
- Visualized and communicated your insights

Next steps

- [Data-driven Decision Making for Business](#)
- [Introduction to Statistics](#)
- [Understanding Data Visualization](#)
- [Data Storytelling Concepts](#)

Congratulations!

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